Artificial Intelligence in the Contemporary World

Project Synopsis

A Collaborative Initiative — Equipping Faculty with Practical Skills in AI, Data Analytics, and Business Intelligence

Project Title	**Edu Pulse AI: Turning Student Voices into Institutional Growth**
Domain	Educational Analytics & NLP
Tools Used	Excel + Power BI + Copilot / AI Chatbot

3. Problem Statement

Most higher education institutions collect student feedback but fail to **convert it into actionable insights that drive measurable improvements in teaching quality, student retention, and institutional reputation**.

Key challenges identified in research:

- **Low accuracy in sentiment analysis** for multilingual and nuanced feedback (especially sarcasm, mixed-language comments).
- **Weak linkage between feedback and teaching strategies** — existing tools stop at reporting sentiment without suggesting interventions.
- **Faculty comparison dashboards** often lack contextual depth, making it hard for institutions to identify best practices across departments.

This project addresses these gaps by building a **business-focused AI feedback analyzer** that transforms raw feedback into **strategic insights for faculty development, accreditation, and student success metrics**.

Research Gaps Identified (Focus Areas for Innovation)

- **Multilingual sentiment accuracy** still below 80% in many studies.
- **Sarcasm and context detection** remains unsolved in student feedback NLP.
- **Actionable linkage to teaching strategies** is rarely implemented in dashboards.

Students should include in their write-up:

- Context of the education sector and challenges posed by unstructured feedback
- Importance of feedback analysis for teaching quality and student success
- The disconnect between feedback collection and actionable insights

- Beneficiaries: faculty, academic coordinators, and institutions

4. Objectives

List the specific goals your project aims to accomplish. Objectives should be clear and measurable.

1. **Consolidate feedback** from ratings, comments, and suggestions into a unified repository.
2. **Enhance multilingual sentiment accuracy** using NLP + Copilot prompt engineering.
3. **Track sentiment trajectories** to measure the impact of teaching interventions over time.
4. **Generate faculty-specific recommendations** (e.g., pacing, clarity, engagement strategies).
5. **Deliver comparative dashboards** for institutional leaders to benchmark departments.
6. **Provide accreditation-ready reports** for audits and curriculum redesign.

5. Methodology / Implementation Plan

The following five-phase workflow provides a structured approach to completing this project.

#	Phase	Description
1	Data Acquisition	Gather feedback datasets from institutional surveys, Kaggle, or simulated records. Include multilingual and mixed-format responses.
2	Preprocessing	Standardise rating scales, anonymise identifiers, translate , and clean text using Power Query.
3	Advanced Analysis	Apply sentiment trajectory analysis, keyword clustering, and topic modelling to uncover evolving student concerns.
4	AI Integration	Use Copilot/Chatbot to generate faculty-specific improvement suggestions and highlight best practices.
5	Visualisation & Delivery0	Build a Power BI dashboard with comparative analytics, sentiment timelines, and actionable insights. Document prompts and outputs.
6	Final Submission	Prepare a detailed report, upload files to GitHub, and record a demonstration video.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop/PC with Intel i3 or higher, 8 GB RAM recommended
- Windows 10/11 or macOS
- Microsoft Excel (2019 or later)
- Power BI Desktop (latest version)
- Microsoft Copilot (via Microsoft 365 subscription)
- AI Chatbot (ChatGPT / Gemini / Copilot Web)
- Internet connection

B. Dataset

- Institutional feedback records or Kaggle datasets
- Multilingual datasets for testing translation and NLP capabilities
- Simulated datasets if required

C. Project Timeline

Phase	Task	Tentative Completion Date
1	Dataset collection & exploration	Week 1
2	Data Dataset collection & exploration	Week 2
3	Sentiment trajectory & topic modelling	Week 2 - 3
4	Dashboard build & AI integration	Week 3
5	Documentation, GitHub upload, video submission	Week 4

7. Expected Outcomes / Conclusion

This section summarises what the team expects to deliver by the end of the project.

- **Improved Teaching Quality** → Faculty receive actionable strategies, boosting student satisfaction scores.
- **Higher Student Retention** → Early detection of dissatisfaction reduces dropout risk.
- **Accreditation Support** → Automated, audit-ready reports streamline compliance.

- **Institutional Reputation Growth** → Data-driven teaching improvements enhance rankings and attract admissions.
- **Scalable Framework** → Reusable across institutions for curriculum reviews and quality audits.

8. References

List all references, tutorials, and documentation used in the project.

- Edunet Foundation. *AICW Fellowship Program Guidelines*. 2026.
- Microsoft Corporation. *Power BI Documentation*.
- Microsoft Corporation. *Microsoft Copilot*.
- Kaggle Inc. *Open Datasets — Student Feedback*.
- LinkedIn Learning. *AI and Data Analytics Courses*. 2026.